

CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 3 (T03) • Set A

Duration **40 minutes** • Maximum marks **20** • 15 questions on four pages

Name _____ SAP ID _____ Batch _____

Date _____ Signature _____ Set

A

Syllabus. Prerequisite material — probability, Python and elementary calculus — together with Units I–III (Lectures 1–3) as retained material: generalisation and data leakage, loss against metric, and batch sizes and momentum; and, principally, **Unit III (Lecture 4):** adaptive optimisers — Adagrad, RMSprop, Adadelata and Adam, bias correction, and what adaptivity does and does not buy.

Instructions. Closed book. Write your answers on this paper, in the space provided. A calculator is not needed — the arithmetic is deliberately short. Answer every question — there is no negative marking. In Part C a reason is **optional**, but a correct reason can rescue a wrong answer. In Parts D and E the marks are for the reasoning, not the vocabulary; an answer that states what it cannot determine earns credit, one that asserts without a reason does not.

About the set letter. Sets are handed out **at random**. Every set covers the same topics at the same level, for the same marks, against the same marking scheme.

Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

- A1.** One card is drawn at random from a standard 52-card pack. Let H be the event that it is a heart. Then $P(H)$ equals
 (a) $1/13$ (b) $1/2$
 (c) $1/4$ (d) $1/52$
- A2.** `len(range(1, 20, 4))` evaluates to
 (a) 5 (b) 4
 (c) 6 (d) an error
- A3.** With n training examples and batch size B , the number of parameter updates in one epoch is
 (a) n (b) B
 (c) 1, whatever B is (d) n/B
- A4.** A step that *learns* from the data is fitted on the whole of X before X is split. The consequence is
 (a) the model underfits (b) information from the test rows influences training
 (c) the optimiser converges more slowly (d) nothing — the split is still random
- A5.** In Adagrad, $s_t = s_{t-1} + g_t \odot g_t$. The quantity s_t is therefore
 (a) a running *average* of squared gradients (b) the running sum of the gradients themselves
 (c) the running *sum* of squared gradients (d) the current gradient, squared

Part B — Fill in the blanks

$4 \times 1 = 4$ marks. Write the missing word on the rule.

- B1.** The derivative of w^3 with respect to w is _____.
- B2.** Averaging the gradient over a mini-batch of size B reduces its noise by a factor of _____.
- B3.** Adagrad's accumulator never decreases, so its effective step size decays like $1/\text{_____}$.
- B4.** Because m and v are corrected for their start at zero, Adam's first step has size exactly _____.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one. A reason is optional, but write one — it can rescue a wrong answer.

C1. Data leakage announces itself through a *poor* score on the held-out data. **TRUE** / **FALSE**

Reason: _____

C2. With the learning rate tuned separately for each, an adaptive optimiser will beat plain gradient descent with momentum on any problem. **TRUE** / **FALSE**

Reason: _____

C3. On a ravine-shaped loss surface, plain momentum can converge in fewer iterations than Adam. **TRUE** / **FALSE**

Reason: _____

Part D — Short answer

2 questions $\times$ 2 = 4 marks. Show how you decide, not just what you decide.

D1. Both optimisers start from $s_0 = 0$ and take their *first* step with $\eta = 0.5$ and gradient $g_1 = -6$. Adagrad uses $s_t = s_{t-1} + g_t^2$; RMSprop uses $s_t = \gamma s_{t-1} + (1 - \gamma)g_t^2$ with $\gamma = 0.9$. Both then apply $w \leftarrow w - \eta g_t / \sqrt{s_t}$; take ε as negligible.

- (a) Compute the size of each optimiser's first step, $|\Delta w_1|$. Give two decimal places where the answer is not exact.
- (b) Neither answer depends on how large g_1 was. Show why.

Part D — Short answer, continued

- D2.** A classifier is trained by minimising cross-entropy, but the figure reported to the client is accuracy. **(a)** Explain why these are two different quantities rather than two names for one. **(b)** State which of them you use to decide the model is finished, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

- E1.** A classmate asserts: “*Adam always beats plain gradient descent with momentum.*” You have one afternoon and a laptop. Design the smallest experiment that would actually settle it.

Answer all four parts overleaf. There is no single right experiment — you are marked on whether the one you describe could actually decide the question.

- (i) What would you run it on? Say how many problems and of what kind. [1]
- (ii) What would you measure, and at what point? [1]
- (iii) Name the one thing that must be held equal between the two optimisers, without which the comparison means nothing at all. [1]
- (iv) State the result that would make you accept the claim, *and* the result that would make you reject it. [1]

Part E — Reasoning, answer space Label each part (i), (ii), (iii). A sketch, a table or a list is as acceptable as prose.

Continue on a continuation sheet if you need more room; label anything you write there with its question number.

CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 3 (T03) • Set B

Duration **40 minutes** • Maximum marks **20** • 15 questions on four pages

Name _____ SAP ID _____ Batch _____

Date _____ Signature _____ Set

B

Syllabus. Prerequisite material — probability, Python and elementary calculus — together with Units I–III (Lectures 1–3) as retained material: generalisation and data leakage, loss against metric, and batch sizes and momentum; and, principally, **Unit III (Lecture 4):** adaptive optimisers — Adagrad, RMSprop, Adadelata and Adam, bias correction, and what adaptivity does and does not buy.

Instructions. Closed book. Write your answers on this paper, in the space provided. A calculator is not needed — the arithmetic is deliberately short. Answer every question — there is no negative marking. In Part C a reason is **optional**, but a correct reason can rescue a wrong answer. In Parts D and E the marks are for the reasoning, not the vocabulary; an answer that states what it cannot determine earns credit, one that asserts without a reason does not.

About the set letter. Sets are handed out **at random**. Every set covers the same topics at the same level, for the same marks, against the same marking scheme.

Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

- A1.** A crate holds 20 items, of which 5 are marked for rework. One is drawn at random. The probability that it is marked is
(a) $1/5$ (b) $1/20$
(c) $3/4$ (d) $1/4$
- A2.** For $xs = [2, 4, 6, 8, 10]$, the expression `len(xs[1:4])` evaluates to
(a) 2 (b) 3
(c) 4 (d) 5
- A3.** A table records $n = 4000$ rows and $B = 32$. The number of parameter updates in one epoch is
(a) 125 (b) 4000
(c) 32 (d) 1
- A4.** A table shows a scaler fitted on all 500 rows, then a 400/100 split, then a test score of 0.99. The most likely reading of that 0.99 is
(a) the model is genuinely excellent (b) the model has underfitted
(c) it is optimistic — test information reached the model (d) the test set is too large
- A5.** Two accumulators are tabulated at each step: one column only ever grows, the other rises and falls. The column that only grows belongs to
(a) RMSprop (b) Adadelata
(c) Adam (d) Adagrad

Part B — Fill in the blanks

$4 \times 1 = 4$ marks. Write the missing word on the rule.

- B1.** A table of w against w^3 has gradient column _____.
- B2.** Measured on the diabetes data, the RMS gradient error fell from 6.32 at $B = 1$ to 1.06 at $B = 32$ — a factor of 5.9, because the noise falls like $1/\text{_____}$.
- B3.** A column of Adagrad's effective step size falls by $9.4\times$ over 2800 updates and keeps falling, because the step decays like $1/\text{_____}$.
- B4.** A table of Adam's first few steps reads 0.1000, 0.0999, 0.0997 at $\eta = 0.1$: the corrected first step has size exactly _____.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one. A reason is optional, but write one — it can rescue a wrong answer.

C1. A results table in which the held-out score is *worse* than expected is the signature of data leakage. **TRUE** / **FALSE**

Reason: _____

C2. A table comparing six optimisers, each at its own tuned learning rate, must show an adaptive method at the top. **TRUE** / **FALSE**

Reason: _____

C3. A table reading 135 iterations for momentum against 262 for Adam on a ravine is possible. **TRUE** / **FALSE**

Reason: _____

Part D — Short answer

2 questions $\times$ 2 = 4 marks. Show how you decide, not just what you decide.

D1. The table below is to be completed for the *first* step of each optimiser from $s_0 = 0$, with $\eta = 0.2$, $\gamma = 0.75$ and $g_1 = +4$. Adagrad accumulates $s_t = s_{t-1} + g_t^2$; RMSprop accumulates $s_t = \gamma s_{t-1} + (1 - \gamma)g_t^2$; both then apply $w \leftarrow w - \eta g_t / \sqrt{s_t}$, with ε negligible.

Optimiser	s_1	$ \Delta w_1 $
Adagrad		
RMSprop		

- (a) Compute the size of each optimiser's first step, $|\Delta w_1|$. Give two decimal places where the answer is not exact.
- (b) Neither answer depends on how large g_1 was. Show why.

Part D — Short answer, continued

- D2.** A results table carries two columns for the same classifier: a cross-entropy column, which the training loop minimised, and an accuracy column, which is what the client was shown. **(a)** Explain why these are two different quantities rather than two names for one. **(b)** State which of them you use to decide the model is finished, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

- E1.** A results table circulating in the lab is being used to argue that “*Adam always beats plain gradient descent with momentum*”. You have one afternoon and a laptop. Design the smallest experiment that would actually settle it.

Answer all four parts overleaf. There is no single right experiment — you are marked on whether the one you describe could actually decide the question.

- (i) What would you run it on? Say how many problems and of what kind. [1]
- (ii) What would you measure, and at what point? [1]
- (iii) Name the one thing that must be held equal between the two optimisers, without which the comparison means nothing at all. [1]
- (iv) State the result that would make you accept the claim, *and* the result that would make you reject it. [1]

Part E — Reasoning, answer space Label each part (i), (ii), (iii). A sketch, a table or a list is as acceptable as prose.

Continue on a continuation sheet if you need more room; label anything you write there with its question number.

CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 3 (T03) • Set C

Duration **40 minutes** • Maximum marks **20** • 15 questions on four pages

Name _____ SAP ID _____ Batch _____

Date _____ Signature _____ Set

C

Syllabus. Prerequisite material — probability, Python and elementary calculus — together with Units I–III (Lectures 1–3) as retained material: generalisation and data leakage, loss against metric, and batch sizes and momentum; and, principally, **Unit III (Lecture 4)**: adaptive optimisers — Adagrad, RMSprop, Adadelat and Adam, bias correction, and what adaptivity does and does not buy.

Instructions. Closed book. Write your answers on this paper, in the space provided. A calculator is not needed — the arithmetic is deliberately short. Answer every question — there is no negative marking. In Part C a reason is **optional**, but a correct reason can rescue a wrong answer. In Parts D and E the marks are for the reasoning, not the vocabulary; an answer that states what it cannot determine earns credit, one that asserts without a reason does not.

About the set letter. Sets are handed out **at random**. Every set covers the same topics at the same level, for the same marks, against the same marking scheme.

Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

- A1.** A spinner is divided into eight equal sectors, two of them shaded. The probability that one spin lands on a shaded sector is
(a) $1/4$ (b) $1/8$
(c) $1/2$ (d) $3/4$
- A2.** `len(set('banana'))` evaluates to
(a) 6 (b) 2
(c) 3 (d) an error
- A3.** A bar chart shows the number of weight updates in one epoch, for three batch sizes run on the same 4000-row dataset. The bar for $B = 32$ has height
(a) 4000 (b) 125
(c) 32 (d) 1
- A4.** A learning curve rises smoothly to a test score far above anything the domain expert expects. Before believing it, the first thing to suspect is
(a) too small a learning rate (b) too large a batch size
(c) an unlucky random seed (d) a fitted step that saw the whole dataset before the split
- A5.** A plot of the denominator $\sqrt{s_t}$ against t is monotonically increasing and never comes back down. The optimiser is
(a) Adagrad (b) RMSprop
(c) Adadelat (d) Adam

Part B — Fill in the blanks

4 $\times$ 1 = 4 marks. Write the missing word on the rule.

- B1.** The slope of the curve $y = w^3$ at a point w is _____.
- B2.** A plot of gradient noise against batch size falls as a power law with exponent $-1/2$: the noise goes like $1/\text{_____}$.
- B3.** A plot of Adagrad's effective step against t is a straight line of slope $-1/2$ on log-log axes, because the step decays like $1/\text{_____}$.
- B4.** A plot of Adam's step size against t starts flat at the value η rather than above it, because bias correction makes the first step exactly _____.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one. A reason is optional, but write one — it can rescue a wrong answer.

C1. On a plot of held-out score against epoch, data leakage shows up as a curve sitting *below* where it should. **TRUE** / **FALSE**

Reason: _____

C2. If six optimisers are each plotted at their own best learning rate, the adaptive ones must end up below the non-adaptive ones. **TRUE** / **FALSE**

Reason: _____

C3. A trajectory plot in which momentum reaches the minimum of a narrow valley in fewer steps than Adam is possible. **TRUE** / **FALSE**

Reason: _____

Part D — Short answer

2 questions $\times 2 = 4$ marks. Show how you decide, not just what you decide.

D1. Two optimisers are to be drawn taking their *first* step from the same point, with $s_0 = 0$, $\eta = 0.4$ and a gradient of $g_1 = -10$ at that point. Adagrad accumulates $s_t = s_{t-1} + g_t^2$; RMSprop accumulates $s_t = \gamma s_{t-1} + (1 - \gamma)g_t^2$ with $\gamma = 0.84$. Both then apply $w \leftarrow w - \eta g_t / \sqrt{s_t}$; take ε as negligible.

- (a) Compute the size of each optimiser's first step, $|\Delta w_1|$. Give two decimal places where the answer is not exact.
- (b) Neither answer depends on how large g_1 was. Show why.

Part D — Short answer, continued

- D2.** Two curves are plotted against the epoch: the cross-entropy the training loop was minimising, and the accuracy that was reported to the client. They do not have the same shape. **(a)** Explain why these are two different quantities rather than two names for one. **(b)** State which of them you use to decide the model is finished, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

- E1.** A plot is being passed around as proof that “*Adam always beats plain gradient descent with momentum*”. You have one afternoon and a laptop. Design the smallest experiment that would actually settle it.

Answer all four parts overleaf. There is no single right experiment — you are marked on whether the one you describe could actually decide the question.

- (i) What would you run it on? Say how many problems and of what kind. [1]
- (ii) What would you measure, and at what point? [1]
- (iii) Name the one thing that must be held equal between the two optimisers, without which the comparison means nothing at all. [1]
- (iv) State the result that would make you accept the claim, *and* the result that would make you reject it. [1]

Part E — Reasoning, answer space Label each part (i), (ii), (iii). A sketch, a table or a list is as acceptable as prose.

Continue on a continuation sheet if you need more room; label anything you write there with its question number.

CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 3 (T03) • Set D

Duration **40 minutes** • Maximum marks **20** • 15 questions on four pages

Name _____ SAP ID _____ Batch _____

Date _____ Signature _____ Set

D

Syllabus. Prerequisite material — probability, Python and elementary calculus — together with Units I–III (Lectures 1–3) as retained material: generalisation and data leakage, loss against metric, and batch sizes and momentum; and, principally, **Unit III (Lecture 4):** adaptive optimisers — Adagrad, RMSprop, Adadelat and Adam, bias correction, and what adaptivity does and does not buy.

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About the set letter. Sets are handed out **at random**. Every set covers the same topics at the same level, for the same marks, against the same marking scheme.

Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

- A1.** A class of 40 students includes 10 who cycle to campus. One student is picked at random. The probability that this student cycles is
(a) $1/10$ (b) $1/4$
(c) $1/40$ (d) $3/4$
- A2.** A shop's price list is written as `{'tea': 20, 'milk': 30, 'tea': 25}`. Its `len` is
(a) 3 (b) 1
(c) an error (d) 2
- A3.** An engineer has 4000 training rows and processes them 32 at a time. In one pass through the data the model's weights are updated
(a) 4000 times (b) 32 times
(c) 125 times (d) once
- A4.** An intern reports a test accuracy far above what the team expected. Asked how, she says she scaled the whole table before splitting it. The reported figure is
(a) optimistic — the test rows helped fit the scaler (b) trustworthy — scaling is not a model
(c) pessimistic — scaling loses information (d) unaffected — scaling is reversible
- A5.** A colleague wants an optimiser that needs no learning rate at all, because the one he picks is always wrong. He wants
(a) Adagrad (b) Adadelat
(c) RMSprop (d) Adam

Part B — Fill in the blanks

4 $\times$ 1 = 4 marks. Write the missing word on the rule.

- B1.** A cost grows as the cube of a setting w . Its rate of change with w is _____.
- B2.** An engineer who quadruples the batch size should expect the gradient noise to halve, because noise falls like $1/\text{_____}$.
- B3.** A colleague complains that after a long run his model has almost stopped moving. He is using Adagrad, whose step decays like $1/\text{_____}$.
- B4.** An engineer notes that whatever the first gradient happens to be, Adam's first move has size exactly _____.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one. A reason is optional, but write one — it can rescue a wrong answer.

C1. A team whose held-out score came out disappointingly low should suspect data leakage. **TRUE** / **FALSE**

Reason: _____

C2. A manager who has tuned the learning rate for every optimiser on the shortlist can be confident the adaptive ones will win. **TRUE** / **FALSE**

Reason: _____

C3. An engineer working on a long narrow valley may find plain momentum reaches the bottom in fewer iterations than Adam. **TRUE** / **FALSE**

Reason: _____

Part D — Short answer

2 questions $\times$ 2 = 4 marks. Show how you decide, not just what you decide.

D1. An engineer runs the same model twice from the same starting weights, once with Adagrad and once with RMSprop, and compares only the *first* step. Both start from $s_0 = 0$ with $\eta = 0.25$, and the gradient at the starting point is $g_1 = +2$. Adagrad accumulates $s_t = s_{t-1} + g_t^2$; RMSprop accumulates $s_t = \gamma s_{t-1} + (1 - \gamma)g_t^2$ with $\gamma = 0.9375$. Both apply $w \leftarrow w - \eta g_t / \sqrt{s_t}$; take ε as negligible.

- (a) Compute the size of each optimiser's first step, $|\Delta w_1|$. Give two decimal places where the answer is not exact.
- (b) Neither answer depends on how large g_1 was. Show why.

Part D — Short answer, continued

- D2.** A client has been shown an accuracy figure for a spam filter. The training loop that produced the filter never used accuracy at all — it minimised cross-entropy. **(a)** Explain why these are two different quantities rather than two names for one. **(b)** State which of them you use to decide the model is finished, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

- E1.** A colleague insists, in a design meeting, that “*Adam always beats plain gradient descent with momentum*” and wants it written into the team’s defaults. You have one afternoon and a laptop. Design the smallest experiment that would actually settle it.

Answer all four parts overleaf. There is no single right experiment — you are marked on whether the one you describe could actually decide the question.

- (i) What would you run it on? Say how many problems and of what kind. [1]
- (ii) What would you measure, and at what point? [1]
- (iii) Name the one thing that must be held equal between the two optimisers, without which the comparison means nothing at all. [1]
- (iv) State the result that would make you accept the claim, *and* the result that would make you reject it. [1]

Part E — Reasoning, answer space Label each part (i), (ii), (iii). A sketch, a table or a list is as acceptable as prose.

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CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 3 (T03) • Set E

Duration **40 minutes** • Maximum marks **20** • 15 questions on four pages

Name _____ SAP ID _____ Batch _____

Date _____ Signature _____ Set

E

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Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

A1. `sum(1 for c in deck if c.suit == 'H') / len(deck)`, for a standard 52-card deck, evaluates to

- (a) 0.0769 (b) 0.5
(c) 0.25 (d) 13

A2. After `xs = [x for x in range(12) if x % 4 == 0]`, `len(xs)` is

- (a) 4 (b) 3
(c) 2 (d) 12

A3. In `for xb, yb in loader: with len(dataset) == 4000 and batch_size=32`, the body runs per epoch

- (a) 125 times (b) 4000 times
(c) 32 times (d) once

A4. `Xs = StandardScaler().fit_transform(X)` is executed before `train_test_split(Xs, y)`. The reported test score is

- (a) unaffected — scaling is deterministic (b) pessimistic — scaling discards information
(c) an error — the shapes will not match (d) optimistic — the test rows helped fit the scaler

A5. In `s = s + g**2` versus `s = gamma*s + (1-gamma)*g**2`, the second line is the update used by

- (a) Adagrad (b) plain SGD
(c) momentum (d) RMSprop

Part B — Fill in the blanks

4 $\times$ 1 = 4 marks. Write the missing word on the rule.

B1. `f = lambda w: w**3`. Then $f'(w)$ is _____.

B2. `g.std()` over mini-batches falls as `batch_size` grows, like $1/\text{_____}$.

B3. With `s += g**2` the denominator `np.sqrt(s)` only ever grows, so the step decays like $1/\text{_____}$.

B4. With `m_hat = m/(1-b1**t)` and `v_hat = v/(1-b2**t)`, the size of Adam's first step is exactly _____.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one. A reason is optional, but write one — it can rescue a wrong answer.

C1. `model.score(X_test, y_test)` coming back *lower* than expected is the signature of data leakage. **TRUE** / **FALSE**

Reason: _____

C2. With `lr` tuned for each, `Adam` must reach a lower final loss than `SGD(momentum=0.9)`. **TRUE** / **FALSE**

Reason: _____

C3. On a badly conditioned quadratic, `SGD(momentum=0.9)` can need fewer iterations than `Adam`. **TRUE** / **FALSE**

Reason: _____

Part D — Short answer

2 questions $\times$ 2 = 4 marks. Show how you decide, not just what you decide.

D1. The two loops below are each run for exactly one step from `s = 0.0`, with `eta = 0.1` and a first gradient of `g = -5.0`; `gamma = 0.96` and `eps` is negligible.

<code># Adagrad</code>	<code># RMSprop</code>
<code>s = s + g**2</code>	<code>s = gamma*s + (1 - gamma)*g**2</code>
<code>w = w - eta*g/np.sqrt(s)</code>	<code>w = w - eta*g/np.sqrt(s)</code>

- Compute the size of each optimiser's first step, $|\Delta w_1|$. Give two decimal places where the answer is not exact.
- Neither answer depends on how large g_1 was. Show why.

Part D — Short answer, continued

D2. The training loop minimises `nn.CrossEntropyLoss()`, but the number in the report comes from `accuracy_score(...)`. **(a)** Explain why these are two different quantities rather than two names for one. **(b)** State which of them you use to decide the model is finished, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

E1. A pull request changes the project default from `SGD(momentum=0.9)` to `Adam`, with the justification “*Adam always beats plain gradient descent with momentum*”. You have one afternoon and a laptop. Design the smallest experiment that would actually settle it.

Answer all four parts overleaf. There is no single right experiment — you are marked on whether the one you describe could actually decide the question.

- (i) What would you run it on? Say how many problems and of what kind. [1]
- (ii) What would you measure, and at what point? [1]
- (iii) Name the one thing that must be held equal between the two optimisers, without which the comparison means nothing at all. [1]
- (iv) State the result that would make you accept the claim, *and* the result that would make you reject it. [1]

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CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 3 (T03) • Set F

Duration **40 minutes** • Maximum marks **20** • 15 questions on four pages

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Date _____ Signature _____ Set

F

Syllabus. Prerequisite material — probability, Python and elementary calculus — together with Units I–III (Lectures 1–3) as retained material: generalisation and data leakage, loss against metric, and batch sizes and momentum; and, principally, **Unit III (Lecture 4):** adaptive optimisers — Adagrad, RMSprop, Adadelata and Adam, bias correction, and what adaptivity does and does not buy.

Instructions. Closed book. Write your answers on this paper, in the space provided. A calculator is not needed — the arithmetic is deliberately short. Answer every question — there is no negative marking. In Part C a reason is **optional**, but a correct reason can rescue a wrong answer. In Parts D and E the marks are for the reasoning, not the vocabulary; an answer that states what it cannot determine earns credit, one that asserts without a reason does not.

About the set letter. Sets are handed out **at random**. Every set covers the same topics at the same level, for the same marks, against the same marking scheme.

Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

- A1.** One card is taken at random from an ordinary pack. The chance that it is a heart is
 (a) one in thirteen (b) one in two
 (c) one in fifty-two (d) one in four
- A2.** Counting the whole numbers from one to nineteen that leave the same remainder as one when divided by four, the count is
 (a) 4 (b) 6
 (c) 5 (d) none of them do
- A3.** Four thousand training examples are worked through thirty-two at a time. In one complete pass, the number of times the weights change is
 (a) four thousand (b) one hundred and twenty-five
 (c) thirty-two (d) once
- A4.** Something that learns from the data is allowed to see all of it before any of it is set aside for testing. The score that comes out afterwards is
 (a) better than it should be (b) worse than it should be
 (c) unchanged (d) impossible to compute
- A5.** Of the four adaptive methods, the one that asks for no step size at all, working one out for itself from the sizes of the steps it has already taken, is
 (a) Adadelata (b) Adagrad
 (c) RMSprop (d) Adam

Part B — Fill in the blanks

4 $\times$ 1 = 4 marks. Write the missing word on the rule.

- B1.** If a quantity grows as the cube of a number, its rate of change with that number is _____.
- B2.** Averaging a gradient over a larger handful of examples makes it less noisy — the noise falls like one over _____.
- B3.** A method whose running total only ever grows must take ever smaller steps: its step falls like one over _____.
- B4.** Correcting both running averages for having started at zero makes the very first step exactly _____ in size, whatever the gradient was.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one. A reason is optional, but write one — it can rescue a wrong answer.

C1. When something that should not have been seen is seen during training, the tell-tale sign is a disappointing score on the data held back. **TRUE / FALSE**

Reason: _____

C2. Once the step size has been chosen carefully for each method in turn, the ones that adapt per coordinate are bound to come out ahead. **TRUE / FALSE**

Reason: _____

C3. On a long narrow valley, carrying momentum can get you to the bottom in fewer steps than a method that adapts each coordinate separately. **TRUE / FALSE**

Reason: _____

Part D — Short answer

2 questions $\times$ 2 = 4 marks. Show how you decide, not just what you decide.

D1. Two methods take their very first step from a standing start, both with a step size of 0.03 and a first gradient of +8. The first keeps a running *total* of squared gradients; the second keeps a running *average* of them, forgetting at a rate that leaves one hundredth of the weight on the newest value. Both then divide the step by the square root of what they have accumulated; the guard constant is too small to matter.

(a) Compute the size of each optimiser's first step, $|\Delta w_1|$. Give two decimal places where the answer is not exact.

(b) Neither answer depends on how large g_1 was. Show why.

Part D — Short answer, continued

- D2.** A spam filter is trained by making one quantity as small as possible, and is then described to the client by a completely different number — the fraction of messages it gets right. **(a)** Explain why these are two different quantities rather than two names for one. **(b)** State which of them you use to decide the model is finished, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

- E1.** Someone claims, flatly, that the method which adapts each coordinate separately always beats the one that merely carries momentum. You have one afternoon and a laptop. Design the smallest experiment that would actually settle it.

Answer all four parts overleaf. There is no single right experiment — you are marked on whether the one you describe could actually decide the question.

- (i) What would you run it on? Say how many problems and of what kind. [1]
- (ii) What would you measure, and at what point? [1]
- (iii) Name the one thing that must be held equal between the two optimisers, without which the comparison means nothing at all. [1]
- (iv) State the result that would make you accept the claim, *and* the result that would make you reject it. [1]

Part E — Reasoning, answer space Label each part (i), (ii), (iii). A sketch, a table or a list is as acceptable as prose.

Continue on a continuation sheet if you need more room; label anything you write there with its question number.

CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 3 (T03) • Set G

Duration **40 minutes** • Maximum marks **20** • 15 questions on four pages

Name _____ SAP ID _____ Batch _____

Date _____ Signature _____ Set

G

Syllabus. Prerequisite material — probability, Python and elementary calculus — together with Units I–III (Lectures 1–3) as retained material: generalisation and data leakage, loss against metric, and batch sizes and momentum; and, principally, **Unit III (Lecture 4):** adaptive optimisers — Adagrad, RMSprop, Adadelata and Adam, bias correction, and what adaptivity does and does not buy.

Instructions. Closed book. Write your answers on this paper, in the space provided. A calculator is not needed — the arithmetic is deliberately short. Answer every question — there is no negative marking. In Part C a reason is **optional**, but a correct reason can rescue a wrong answer. In Parts D and E the marks are for the reasoning, not the vocabulary; an answer that states what it cannot determine earns credit, one that asserts without a reason does not.

About the set letter. Sets are handed out **at random**. Every set covers the same topics at the same level, for the same marks, against the same marking scheme.

Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

- A1.** Compare, for one card drawn from a standard pack, $P(\text{a heart})$ with $P(\text{an ace})$
 (a) the first is larger (b) the second is larger
 (c) they are equal (d) cannot be decided
- A2.** Compare `len(range(1, 20, 4))` with `len('adam')`
 (a) the second is larger (b) they are equal
 (c) one raises an error (d) the first is larger
- A3.** Compare the updates per epoch made by $B = 1$ with those made by $B = 32$, on the same data
 (a) $B = 32$ makes $32\times$ as many (b) they are equal
 (c) $B = 1$ makes $32\times$ as many (d) it depends on η
- A4.** Compare fitting a scaler *inside* each fold with fitting it once on everything beforehand
 (a) the second deflates the reported score (b) the second inflates the reported score
 (c) they give the same score (d) the second is faster and otherwise identical
- A5.** Compare what Adagrad accumulates with what RMSprop accumulates
 (a) an average, against a sum (b) a sum, against an average
 (c) both sums, at different rates (d) both averages, at different rates

Part B — Fill in the blanks

$4 \times 1 = 4$ marks. Write the missing word on the rule.

- B1.** Compare w^3 with its derivative: the second is _____.
- B2.** Compare the gradient noise at batch size B with that at batch size 1: the first is smaller by a factor of _____.
- B3.** Compare Adagrad's step at step t with its step at step 1: it has shrunk like $1/\text{_____}$.
- B4.** Compare Adam's corrected first step with its uncorrected one: the corrected version has size exactly _____.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one. A reason is optional, but write one — it can rescue a wrong answer.

C1. Between a model with leakage and one without, it is the leaking model that reports the *lower* held-out score.
TRUE / FALSE

Reason: _____

C2. Between an adaptive optimiser and momentum, each at its own tuned learning rate, the adaptive one always finishes lower. **TRUE / FALSE**

Reason: _____

C3. Between momentum and Adam on a ravine, it can be momentum that needs fewer iterations. **TRUE / FALSE**

Reason: _____

Part D — Short answer

2 questions $\times$ 2 = 4 marks. Show how you decide, not just what you decide.

D1. Compare the *first* step taken by Adagrad with the first step taken by RMSprop, both from $s_0 = 0$, both with $\eta = 0.6$, and both at a point where the gradient is $g_1 = -3$. Adagrad accumulates $s_t = s_{t-1} + g_t^2$; RMSprop accumulates $s_t = \gamma s_{t-1} + (1 - \gamma)g_t^2$ with $\gamma = 0.8$. Both apply $w \leftarrow w - \eta g_t / \sqrt{s_t}$; take ε as negligible.

- (a) Compute the size of each optimiser's first step, $|\Delta w_1|$. Give two decimal places where the answer is not exact.
- (b) Neither answer depends on how large g_1 was. Show why.

Part D — Short answer, continued

- D2.** Compare the two numbers attached to one classifier: the cross-entropy its training loop minimised, and the accuracy quoted to the client. **(a)** Explain why these are two different quantities rather than two names for one. **(b)** State which of them you use to decide the model is finished, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

- E1.** Two colleagues disagree. One says “*Adam always beats plain gradient descent with momentum*”; the other says it depends. You have one afternoon and a laptop. Design the smallest experiment that would actually settle it between them.

Answer all four parts overleaf. There is no single right experiment — you are marked on whether the one you describe could actually decide the question.

- (i) What would you run it on? Say how many problems and of what kind. [1]
- (ii) What would you measure, and at what point? [1]
- (iii) Name the one thing that must be held equal between the two optimisers, without which the comparison means nothing at all. [1]
- (iv) State the result that would make you accept the claim, *and* the result that would make you reject it. [1]

Part E — Reasoning, answer space Label each part (i), (ii), (iii). A sketch, a table or a list is as acceptable as prose.

Continue on a continuation sheet if you need more room; label anything you write there with its question number.

CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 3 (T03) • Set H

Duration **40 minutes** • Maximum marks **20** • 15 questions on four pages

Name _____ SAP ID _____ Batch _____

Date _____ Signature _____ Set

H

Syllabus. Prerequisite material — probability, Python and elementary calculus — together with Units I–III (Lectures 1–3) as retained material: generalisation and data leakage, loss against metric, and batch sizes and momentum; and, principally, **Unit III (Lecture 4):** adaptive optimisers — Adagrad, RMSprop, Adadelat and Adam, bias correction, and what adaptivity does and does not buy.

Instructions. Closed book. Write your answers on this paper, in the space provided. A calculator is not needed — the arithmetic is deliberately short. Answer every question — there is no negative marking. In Part C a reason is **optional**, but a correct reason can rescue a wrong answer. In Parts D and E the marks are for the reasoning, not the vocabulary; an answer that states what it cannot determine earns credit, one that asserts without a reason does not.

About the set letter. Sets are handed out **at random**. Every set covers the same topics at the same level, for the same marks, against the same marking scheme.

Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

- A1.** *One card in four is a heart, and $13/52 = 1/4$.* Drawing one card at random from a standard pack, $P(\text{a heart})$ is
 (a) $1/13$ (b) $1/4$
 (c) $1/2$ (d) 13
- A2.** *$\text{len}(\text{range}(1, 10, 3))$ is 3, counting 1, 4, 7.* Then $\text{len}(\text{range}(1, 20, 4))$ is
 (a) 5 (b) 4
 (c) 6 (d) an error
- A3.** *With $n = 100$ and $B = 10$, one epoch makes 10 updates.* With $n = 4000$ and $B = 32$, one epoch makes
 (a) 4000 (b) 32
 (c) 1 (d) 125
- A4.** *A scaler fitted inside each fold gives an honest score.* The same scaler fitted on everything before the split gives a score that is
 (a) too poor (b) identical
 (c) too good (d) undefined
- A5.** *Adagrad keeps a running sum of squared gradients.* RMSprop keeps a running
 (a) sum of squared gradients (b) sum of the gradients themselves
 (c) average of squared gradients (d) average of the gradients themselves

Part B — Fill in the blanks

4 $\times$ 1 = 4 marks. Write the missing word on the rule.

- B1.** *The derivative of w^2 is $2w$.* The derivative of w^3 is _____.
- B2.** *Going from $B = 1$ to $B = 4$ halves the gradient noise.* In general the noise falls like $1/\text{_____}$.
- B3.** *After 100 steps Adagrad's step has shrunk tenfold.* In general the step decays like $1/\text{_____}$.
- B4.** *Uncorrected, Adam's first step is $3.16\times$ too large at $\beta_2 = 0.999$.* Corrected, its size is exactly _____.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one. A reason is optional, but write one — it can rescue a wrong answer.

C1. *An honest pipeline gives a believable held-out score. A leaking one therefore gives a score that is believably low.*
TRUE / FALSE

Reason: _____

C2. *On a sparse problem, Adagrad beat plain SGD by a factor of eight. Therefore an adaptive optimiser, tuned, beats momentum on any problem.* **TRUE / FALSE**

Reason: _____

C3. *Adaptive methods measure gradients, and a ravine is a problem of curvature. So on a ravine, momentum can beat Adam.* **TRUE / FALSE**

Reason: _____

Part D — Short answer

2 questions $\times$ 2 = 4 marks. Show how you decide, not just what you decide.

D1. *From $s_0 = 0$ with $\eta = 1$ and $g_1 = -2$, Adagrad has $s_1 = 4$, so its first step is $1 \times (-2)/2 = -1$: size 1, which is η . Now take $\eta = 0.05$ and $g_1 = +12$, with RMSprop's $\gamma = 0.95$. Adagrad accumulates $s_t = s_{t-1} + g_t^2$; RMSprop accumulates $s_t = \gamma s_{t-1} + (1 - \gamma)g_t^2$. Both apply $w \leftarrow w - \eta g_t / \sqrt{s_t}$; take ε as negligible.*

(a) Compute the size of each optimiser's first step, $|\Delta w_1|$. Give two decimal places where the answer is not exact.

(b) Neither answer depends on how large g_1 was. Show why.

Part D — Short answer, continued

D2. *A ruler and a thermometer both produce numbers, but they are not two names for one measurement.* A classifier's cross-entropy and its accuracy are likewise two different things. **(a)** Explain why. **(b)** State which of them you use to decide the model is finished, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

E1. *Worked analogue. To settle “drug A cures faster than drug B” you would not give A to the fit patients and B to the frail ones — you would hold everything but the drug equal, and decide in advance what difference would count.*

Now settle this one. A classmate asserts: “*Adam always beats plain gradient descent with momentum.*” You have one afternoon and a laptop. Design the smallest experiment that would actually settle it.

Answer all four parts overleaf. There is no single right experiment — you are marked on whether the one you describe could actually decide the question.

- (i) What would you run it on? Say how many problems and of what kind. [1]
- (ii) What would you measure, and at what point? [1]
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Continue on a continuation sheet if you need more room; label anything you write there with its question number.